HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination — Set C (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

— Version with detailed solutions —

Duration: **120 minutes**
Total credit: **120 points**
Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: ______________________________ Matriculation no.: ____________________

Problem	P1	P2	P3	P4	P5	P6	P7	Total
Maximum points	16	8	20	16	20	20	20	**120**
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e, logarithms, $(0.95)^{40}$, t-quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines (16 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[4 P]** A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at each knot. Derive the number of degrees of freedom (free parameters, counting the intercept) of a cubic spline with K interior knots by counting the coefficients of the polynomial pieces and subtracting the constraints imposed at the knots. Evaluate your formula for $K = 3$ interior knots.

(b) **[4 P]** How many degrees of freedom does a *natural* cubic spline with the same $K = 3$ knots use? State the additional constraint a natural spline imposes, say in which region of the predictor space it takes effect, and explain why it is useful in practice.

(c) **[4 P]** A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^{n}\bigl(y_i - g(x_i)\bigr)^2 + \lambda \int g''(t)^2\,dt.$$

Explain the role of each of the two terms, and describe the behaviour of the solution $\hat{g}$ in the two limiting cases $\lambda \to 0$ and $\lambda \to \infty$.

(d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ, define the truncated-power basis function explicitly, and explain in one sentence why adding this basis function does not destroy the smoothness of the fit at ξ.

Solution

(a) Degrees of freedom of a cubic spline. K interior knots split the range of X into $K+1$ intervals, each carrying a cubic polynomial with 4 coefficients: $4(K+1) = 4K+4$ parameters. At each knot, continuity of g, g' and g'' imposes 3 constraints, i.e. $3K$ constraints in total. Hence

$$\mathrm{df} = 4(K+1) - 3K = K+4.$$

For $K = 3$: $\mathrm{df} = 3+4 = \mathbf{7}$ free parameters (equivalently: intercept, x, x^2, x^3 plus one truncated-power basis function per knot $= 4+K$).

(b) Natural cubic spline. A natural spline additionally requires the function to be *linear beyond the boundary knots*, i.e. in the two outer regions $X < \xi_1$ and $X > \xi_K$. Forcing the quadratic and cubic terms to vanish there removes 2 parameters per boundary region, i.e. 4 in total:

$$\mathrm{df} = (K+4) - 4 = K, \qquad K = 3: \ \mathrm{df} = \mathbf{3}.$$

Usefulness: polynomials and unconstrained splines are notoriously erratic near the boundaries, where data are sparse; the linearity constraint stabilises the fit there (*lower variance* of predictions at and beyond the boundary), at the price of a small amount of additional bias.

(c) Smoothing spline. The first term (RSS) rewards fidelity to the data; the second term penalises roughness, since g'' measures the curvature (wiggliness) of g, and $\lambda \geq 0$ is the tuning parameter that trades the two off. *$\lambda \to 0$:* the penalty disappears; $\hat{g}$ can be made arbitrarily wiggly and *interpolates* the training observations exactly (zero training RSS, severe overfitting; effective df $\to n$). *$\lambda \to \infty$:* any curvature is infinitely expensive, so $g'' \equiv 0$ is enforced; $\hat{g}$ becomes a straight line and equals the *least-squares regression line* (effective df $\to 2$). Between the extremes, λ (chosen e.g. by LOOCV) controls the effective degrees of freedom, which decrease smoothly from n to 2.

(d) Truncated-power basis with one knot ξ.

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x-\xi)_+^3, \qquad (x-\xi)_+^3 = \begin{cases} (x-\xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

The function $(x-\xi)_+^3$ and its first and second derivatives are all 0 at $x = \xi$, so adding it changes only the third derivative at the knot; the fit remains continuous with continuous first and second derivatives — exactly the defining smoothness of a cubic spline.

Problem 2: Generalised additive models (8 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \dots, X_p$, and explain in one sentence how it generalises multiple linear regression.

(b) **[3 P]** State one important *advantage* of GAMs concerning the interpretation of the fitted functions f_j, and the main structural *limitation* of the additive form — including how this limitation can be repaired.

(c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Solution

(a) Model form.

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p) + \varepsilon.$$

Multiple linear regression is the special case $f_j(X_j) = \beta_j X_j$; a GAM replaces each linear component by a smooth non-linear function f_j (e.g. a natural or smoothing spline, or local regression), while keeping the *additive* structure.

(b) Advantage and limitation. *Advantage (interpretability):* because the model is additive, the contribution of each X_j to Y is captured by its own function f_j, which can be examined and plotted individually while the other variables are held fixed — non-linear effects remain as easy to read as in a coefficient table, one panel per predictor. *Limitation:* additivity excludes *interactions*: the effect of X_j cannot depend on the value of X_k. Repair: add interaction terms manually, e.g. a product term $X_j \times X_k$ or a low-dimensional smooth surface $f_{jk}(X_j, X_k)$ as an extra component.

(c) Backfitting. Cycle through the predictors and repeatedly re-fit each f_j (by a smoother, e.g. a spline) to the *partial residuals* $r_i = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$, holding all other fitted functions fixed; iterate until the f_j stop changing.

Problem 3: Regression and classification trees by hand (20 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

A logistics company records for $n = 6$ regional depots the number of delivery vans in operation x and the number of parcels delivered that day y (in 1,000 parcels):

Depot i	1	2	3	4	5	6
Vans x_i	1	2	3	4	5	6
Parcels y_i	3	5	4	9	11	10

A regression tree with a *single* split at cutpoint s partitions the data into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

(a) **[8 P]** For each of the two candidate cutpoints $s = 2.5$ and $s = 3.5$, determine the two regions, the region means and the resulting total residual sum of squares $\text{RSS}(s) = \sum_{m=1}^{2} \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting choose, and why?

(b) **[2 P]** Using the chosen split, predict the number of parcels for a new depot that will operate $x = 5$ vans.

(c) **[6 P]** A *classification* tree for a different task has a terminal node containing 9 training observations from three product categories: 5 "Electronics", 3 "Clothing" and 1 "Books". Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node *(use* $\ln(5/9) = -0.5878$*,* $\ln(1/3) = -1.0986$ *and* $\ln(1/9) = -2.1972$*)*. Which class does the node predict?

(d) **[4 P]** In *cost-complexity pruning* one minimises, for a subtree $T \subseteq T_0$ of the large tree T_0,

$$\sum_{m=1}^{|T|} \sum_{i:\, x_i \in R_m} \left(y_i - \hat{y}_{R_m}\right)^2 + \alpha\, |T|.$$

Explain the role of the tuning parameter α: what happens for $\alpha = 0$, what happens as α increases, and how is α chosen in practice?

Solution

(a) Evaluating the two cutpoints. Root node for reference: $\bar{y} = 42/6 = 7$.
Cutpoint $s = 2.5$: $R_1 = \{1, 2\}$ with y-values $\{3, 5\}$, mean $\hat{y}_{R_1} = 4$; $R_2 = \{3, 4, 5, 6\}$ with y-values $\{4, 9, 11, 10\}$, mean $\hat{y}_{R_2} = 34/4 = 8.5$.

$$\begin{aligned} \mathrm{RSS}_{R_1} &= (3-4)^2 + (5-4)^2 = 2, \\ \mathrm{RSS}_{R_2} &= (4-8.5)^2 + (9-8.5)^2 + (11-8.5)^2 + (10-8.5)^2 = 20.25 + 0.25 + 6.25 + 2.25 = 29, \end{aligned}$$

so $\mathrm{RSS}(2.5) = 2 + 29 = \mathbf{31}$.
Cutpoint $s = 3.5$: $R_1 = \{1, 2, 3\}$ with y-values $\{3, 5, 4\}$, mean $\hat{y}_{R_1} = 12/3 = 4$; $R_2 = \{4, 5, 6\}$ with y-values $\{9, 11, 10\}$, mean $\hat{y}_{R_2} = 30/3 = 10$.

$$\mathrm{RSS}_{R_1} = (3-4)^2 + (5-4)^2 + (4-4)^2 = 2, \qquad \mathrm{RSS}_{R_2} = (9-10)^2 + (11-10)^2 + (10-10)^2 = 2,$$

so $\mathrm{RSS}(3.5) = 2 + 2 = \mathbf{4}$.
Since $4 < 31$, recursive binary splitting is *greedy* and picks the cutpoint with the smallest RSS: $\mathbf{s = 3.5}$. (It cleanly separates the three low-volume depots, mean 4, from the three high-volume depots, mean 10; in fact $s = 3.5$ is the best of all five possible cutpoints.)

(b) Prediction for $x = 5$. $x = 5 \geq 3.5$, so the new depot falls into R_2 and the tree predicts $\hat{y} = \hat{y}_{R_2} = \mathbf{10}$ (i.e. 10,000 parcels).

(c) Node impurity (three classes). $\hat{p}_{\mathrm{E}} = 5/9$, $\hat{p}_{\mathrm{C}} = 3/9 = 1/3$, $\hat{p}_{\mathrm{B}} = 1/9$.

$$G = \tfrac{5}{9} \cdot \tfrac{4}{9} + \tfrac{3}{9} \cdot \tfrac{6}{9} + \tfrac{1}{9} \cdot \tfrac{8}{9} = \tfrac{20+18+8}{81} = \tfrac{46}{81} = \mathbf{0.568}.$$

$$\begin{aligned} D &= -\left(\tfrac{5}{9} \ln \tfrac{5}{9} + \tfrac{1}{3} \ln \tfrac{1}{3} + \tfrac{1}{9} \ln \tfrac{1}{9}\right) = -\left(\tfrac{5}{9}(-0.5878) + \tfrac{1}{3}(-1.0986) + \tfrac{1}{9}(-2.1972)\right) \\ &= 0.3266 + 0.3662 + 0.2441 = \mathbf{0.937}. \end{aligned}$$

The node predicts the majority class, **Electronics** (with estimated probability $5/9 \approx 0.556$). Both G and D would be 0 for a pure node; here the node is quite impure because no class dominates strongly.

(d) Role of α. α is the price per terminal node: it trades off the fit of the subtree (first term) against its complexity $|T|$. For $\alpha = 0$ the criterion equals the training RSS and the full tree T_0 is optimal. As α increases, additional leaves must "pay for themselves"; branches whose RSS reduction is smaller than α are pruned, so the optimal subtree becomes smaller — one obtains a *nested sequence* of subtrees as α grows (weakest-link pruning). In practice α is selected by *cross-validation* (choose the α with the smallest CV error, then refit on all data), which balances bias against variance instead of optimising training fit.

Problem 4: Ensembles — bagging, random forests, boosting (16 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

(a) **[6 P]** Let $\hat{f}_1(x), \ldots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x, each with variance σ^2 and with pairwise correlation ρ. The variance of the bagged average is

$$\mathrm{Var}\Big(\tfrac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)\Big) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2.$$

(i) Explain briefly why this formula shows that bagging reduces variance, and which term limits the reduction. (ii) Evaluate the variance for $B = 50$, $\rho = 0.4$ and $\sigma^2 = 16$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \to \infty$?

(b) **[4 P]** How do *random forests* modify bagging at each split, and what is the recommended subset size m for *regression* as a function of p (evaluate for $p = 30$)? Explain, with reference to the formula in (a), why this modification improves on bagging.

(c) **[4 P]** Name the three tuning parameters of *boosting* for regression trees and state in one sentence each what they control. What is the "slow learning" idea behind boosting?

(d) **[2 P]** What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it available essentially for free?

Solution

(a) Variance of the bagged average. (i) The second term $\frac{1-\rho}{B}\sigma^2$ is driven to 0 by averaging over many trees (B large), so averaging removes the "independent part" of the trees' variability — this is the variance reduction of bagging. The first term $\rho\sigma^2$ does *not* depend on B: because the bootstrap trees are grown on overlapping samples of the same data (and share the same strong predictors), they are positively correlated, and this correlated part of the variance cannot be averaged away. It limits the reduction. (ii) For $B = 50$, $\rho = 0.4$, $\sigma^2 = 16$:

$$\mathrm{Var} = 0.4 \cdot 16 + \frac{1-0.4}{50}\cdot 16 = 6.4 + \frac{0.6\cdot 16}{50} = 6.4 + 0.192 = \mathbf{6.592},$$

compared with $\sigma^2 = 16$ for a single tree — a reduction of about 58.8%. (iii) As $B \to \infty$ the variance approaches the floor $\rho\sigma^2 = 0.4 \cdot 16 = \mathbf{6.4}$. (Increasing B further does not overfit; it only stabilises the average.)

(b) Random forests. At each split, only a *random subset* of m of the p predictors is considered as splitting candidates; a fresh subset is drawn at every split. Recommended for regression: $m \approx p/3$, so for $p = 30$: $m = 30/3 = \mathbf{10}$. Why it helps: with bagging, a dominant predictor is used in the top split of almost every tree, so the trees look alike and ρ is high. Excluding it from $\approx (p-m)/p$ of the split searches *decorrelates* the trees, i.e. lowers ρ — and by (a) this lowers the irreducible floor $\rho\sigma^2$ of the ensemble variance, which mere averaging cannot touch.

(c) Boosting.

- *Number of trees B:* how many small trees are added sequentially; unlike bagging, boosting *can* overfit if B is too large, so B is chosen by cross-validation.
- *Shrinkage* λ (learning rate, e.g. 0.01 or 0.001): scales the contribution of each new tree and thus the speed of learning; smaller λ usually needs larger B.
- *Number of splits d per tree* (interaction depth): the complexity of each tree and the interaction order of the model; often even $d = 1$ (stumps, additive model) works well.

Slow learning: each tree is fitted to the current *residuals*, and only a small fraction λ of it is added to the model; the model thus improves slowly and exactly in the regions where it still performs poorly, instead of fitting the data hard in one step — which tends to generalise better.

(d) OOB error. Each bootstrap sample leaves out on average about 1/3 of the observations ("out-of-bag"). Predicting each observation only with the trees that did *not* use it, and aggregating these predictions, yields the OOB error — a valid estimate of the *test error* of the bagged model. It is free because it reuses the bootstrap side product; no separate validation set or cross-validation loop is required.

Problem 5: Neural networks by hand (20 points)

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Consider a tiny feed-forward network for a quantitative response with input $x = (x_1, x_2) = (2, 3)$, one hidden layer with two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g\big(-3 + 2\,x_1 + 1\,x_2\big), \qquad A_2 = g\big(2 - 2\,x_1 - 1\,x_2\big), \qquad f(x) = -2 + 3\,A_1 + 2\,A_2.$$

(a) **[8 P]** Perform the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the network output $f(x)$. Comment briefly on what the ReLU did to the second unit and why such non-linear activations are essential.

(b) **[6 P]** A classification network with $K = 3$ classes produces the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (2.0,\ 0.0,\ -1.0)$ for one observation whose true class is class 2. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. *(Use* $e^2 = 7.389$*,* $e^0 = 1.000$*,* $e^{-1} = 0.368$ *and* $\ln 8.757 = 2.170$*.)*

(c) **[4 P]** Count the parameters of a fully connected network with $p = 6$ inputs, one hidden layer with $k = 5$ units and an output layer with $K = 3$ units, where every hidden and output unit has a bias. Show the count separately for the two weight layers.

(d) **[2 P]** A convolutional layer receives a 64×64 image and uses filters of size $F = 3$ with padding $P = 1$ and stride $S = 2$. Compute the spatial size of the output feature map from $\lfloor (W - F + 2P)/S \rfloor + 1$ (note the stride is 2, so the division is rounded down).

Solution

(a) Forward pass. Pre-activations:

$$z_1 = -3 + 2 \cdot 2 + 1 \cdot 3 = -3 + 4 + 3 = 4, \qquad z_2 = 2 - 2 \cdot 2 - 1 \cdot 3 = 2 - 4 - 3 = -5.$$

ReLU activations:

$$A_1 = g(4) = \max\{0, 4\} = 4, \qquad A_2 = g(-5) = \max\{0, -5\} = 0.$$

Output:

$$f(x) = -2 + 3 \cdot 4 + 2 \cdot 0 = -2 + 12 + 0 = \mathbf{10}.$$

The ReLU *switched the second unit off*: its negative pre-activation is clipped to 0, so unit 2 contributes nothing for this input (but would for other inputs). This thresholding is the source of non-linearity: without a non-linear g, the network would collapse into a single linear function of x, no matter how many layers it has.

(b) Softmax and cross-entropy. Exponentials: $e^{2.0} = 7.389$, $e^{0.0} = 1.000$, $e^{-1.0} = 0.368$; their sum is $7.389 + 1.000 + 0.368 = 8.757$. Softmax probabilities:

$$\hat{p}_1 = \frac{7.389}{8.757} = 0.844, \qquad \hat{p}_2 = \frac{1.000}{8.757} = 0.114, \qquad \hat{p}_3 = \frac{0.368}{8.757} = 0.042,$$

which sum to 1.000. Cross-entropy loss for the true class 2:

$$-\ln \hat{p}_2 = -\ln \frac{1.000}{8.757} = -\big(0 - \ln 8.757\big) = \ln 8.757 = \mathbf{2.170}.$$

(Note that the model's most probable class is class 1, but the loss is evaluated at the *true* class 2, which received only probability 0.114 — hence the sizeable loss.)

(c) Parameter count for $6 \to 5 \to 3$. Input $\to$ hidden: $6 \cdot 5 = 30$ weights $+ 5$ biases $= 35$. Hidden $\to$ output: $5 \cdot 3 = 15$ weights $+ 3$ biases $= 18$. Total: $35 + 18 = \mathbf{53}$ parameters.

(d) Convolution output size.

$$\left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{64 - 3 + 2 \cdot 1}{2} \right\rfloor + 1 = \left\lfloor \frac{63}{2} \right\rfloor + 1 = \lfloor 31.5 \rfloor + 1 = 31 + 1 = 32,$$

so the output feature map is $\mathbf{32} \times \mathbf{32}$ — the stride $S = 2$ roughly halves the 64×64 input (the floor discards the fractional part before adding 1).

Problem 6: Multiple testing (20 points)

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

(a) **[4 P]** A lab performs $m = 40$ *independent* hypothesis tests, each at level $\alpha = 0.05$, and suppose all 40 null hypotheses are true. Define the family-wise error rate (FWER) and compute it for this situation *(use* $(0.95)^{40} = 0.1285$*)*. Interpret the result in one sentence.

(b) **[7 P]** A research group screens $m = 6$ candidate biomarkers; the ordered p-values are

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state for each method exactly which hypotheses are rejected, and explain why Holm can never reject fewer hypotheses than Bonferroni.

(c) **[6 P]** Apply the *Benjamini–Hochberg* (BH) procedure with FDR level $q = 0.05$ to the same six p-values: give the thresholds jq/m, determine which hypotheses are rejected, and state precisely what "FDR control at $q = 0.05$" guarantees for the resulting set of rejections.

(d) **[3 P]** A pharmaceutical company runs a *confirmatory* phase-III trial that tests $m = 5$ pre-specified endpoints; a single false-positive endpoint could lead to the approval of an ineffective drug. Should they control the FWER or the FDR? Justify your recommendation.

Solution

(a) FWER for $m = 40$ independent tests. FWER $= \Pr(\text{at least one Type I error}) = \Pr(V \geq 1)$, where V is the number of true nulls that are (falsely) rejected. With independence and all nulls true:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{40} = 1 - (0.95)^{40} = 1 - 0.1285 = \mathbf{0.8715}.$$

Interpretation: even though each single test is performed at the 5% level, there is about an 87% chance of at least one false discovery among the 40 tests — testing many hypotheses without adjustment almost guarantees spurious findings.

(b) Bonferroni and Holm at $\alpha = 0.05$. *Bonferroni:* reject $H_{(j)}$ iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$. $p_{(1)} = 0.001 \leq 0.00833$ ✓, $p_{(2)} = 0.008 \leq 0.00833$ ✓, $p_{(3)} = 0.011 > 0.00833$ ×. $\Rightarrow$ Bonferroni rejects $H_{(1)}$ and $H_{(2)}$: **2 rejections**.
Holm: compare $p_{(j)}$ with $\alpha/(m - j + 1)$ step by step and stop at the first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
$\alpha/(m-j+1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
reject?	✓	✓	✓	stop	—	—

$0.001 \le 0.00833$, $0.008 \le 0.0100$, $0.011 \le 0.0125$, but $0.030 > 0.0167$: stop. $\Rightarrow$ Holm rejects $H_{(1)}, H_{(2)}, H_{(3)}$: **3 rejections**.
Why Holm $\supseteq$ Bonferroni: Holm's thresholds $\alpha/(m-j+1)$ are $\ge \alpha/m$ for every j (with equality only at $j = 1$), so every p-value that passes Bonferroni also passes Holm's less stringent later hurdles — Holm is uniformly more powerful, yet still controls the FWER at α without any independence assumption. Here Holm rejects one more hypothesis ($H_{(3)}$) than Bonferroni.

(c) Benjamini–Hochberg at $q = 0.05$. Thresholds $jq/m = j \cdot 0.05/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \le jq/m$?	yes	yes	yes	yes	**yes**	no

L = largest j with $p_{(j)} \le jq/m$: $j = 5$ (since $0.041 \le 0.0417$, while $0.260 > 0.0500$). BH rejects *all* $H_{(j)}$ with $j \le L$: $H_{(1)}, \dots, H_{(5)}$: **5 rejections**. Guarantee: $\text{FDR} = E\big[V/\max(R,1)\big] \le q = 0.05$ — *on average* at most 5% of the rejected hypotheses are false discoveries. It does *not* guarantee that at most 5% of these particular 5 rejections are false, nor does it bound the probability of making any false rejection (that would be FWER control). Note BH is the most liberal of the three here (5 rejections vs. 3 for Holm and 2 for Bonferroni).

(d) Confirmatory phase-III trial. Control the **FWER**. This is a confirmatory study with only $m = 5$ pre-specified endpoints, and the cost of even a single false positive is severe (approval of an ineffective drug, patient harm, regulatory failure). FWER control (e.g. Holm at $\alpha = 0.05$) bounds the probability of *any* false rejection at 5%, which is exactly the guarantee a regulator demands. FDR control would only limit the *expected proportion* of false endpoints and would tolerate an occasional false positive — acceptable in large exploratory screens, but not when each individual claim must stand on its own.

Problem 7: Cumulative essentials (Chapters 2–6) (20 points)

Lecture slides: Chapters 2–6 (Lectures 1–8).

(a) **[4 P]** The *bootstrap* draws a resample of size n with replacement from the n observations. (i) Show that the probability that a given observation is *not* selected into a particular bootstrap sample equals $(1 - 1/n)^n$, and give its limit as $n \to \infty$ *(use $e^{-1} = 0.368$)*. (ii) Hence what fraction of the original observations appears (at least once) in a typical bootstrap sample as $n \to \infty$? (iii) Compute this inclusion probability exactly for $n = 5$ *(use $0.8^5 = 0.328$)*.

(b) **[4 P]** At a fixed test point x_0 the expected test error of an estimate $\hat{f}$ decomposes as $E\big[(y_0 - \hat{f}(x_0))^2\big] = \text{Var}(\hat{f}(x_0)) + \big[\text{Bias}(\hat{f}(x_0))\big]^2 + \text{Var}(\varepsilon)$. Two candidate methods give, at x_0:

Method	$\text{Var}(\hat{f})$	$\text{Bias}(\hat{f})$	$\text{Var}(\varepsilon)$
(1) flexible spline	3.5	−1.0	2.0
(2) linear model	0.5	−2.5	2.0

Compute the expected test MSE of each method, say which you would deploy at x_0, and state the lowest test MSE *any* method could achieve here.

(c) **[3 P]** Write down the *elastic net* penalty and explain what it inherits from the lasso and what it inherits from ridge regression. In which situation does the elastic net clearly outperform the lasso?

(d) **[3 P]** Two predictors in a regression have correlation 0.95. Explain what such *collinearity* does to the least-squares coefficient estimates. The variance inflation factor is $\mathrm{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is from regressing X_j on the other predictors; compute and interpret VIF_j for $R_j^2 = 0.9$ *(use $\sqrt{10} = 3.162$).*

(e) **[6 P]** Match each scenario to exactly one method from {*linear regression*, *lasso*, *logistic regression*, *random forest*} (each method used exactly once) and justify each choice in one sentence:

(i) Estimate, with a 95% confidence interval, how much one additional year of schooling raises hourly wage, from $n = 3{,}000$ workers and a handful of predictors.
(ii) From $n = 80$ patients with $p = 8{,}000$ metabolite measurements, predict a continuous biomarker level and identify a short list of relevant metabolites.
(iii) Predict whether a customer will churn (yes/no) and report to management how each customer characteristic changes the *odds* of churning.
(iv) Forecast next-day electricity demand (continuous) from hundreds of weather and calendar variables with strong non-linear effects and interactions; only forecast accuracy matters.

Solution

(a) Bootstrap inclusion probability. (i) In each of the n independent draws, a fixed observation is *missed* with probability $1 - \frac{1}{n}$; since the draws are independent, all n draws miss it with probability $(1 - \frac{1}{n})^n$. As $n \to \infty$, $(1 - \frac{1}{n})^n \to e^{-1} = \mathbf{0.368}$. (ii) The observation is therefore *included* with probability $1 - e^{-1} = 1 - 0.368 = 0.632$, so on average about **63.2**% of the original observations appear in a bootstrap sample (the remaining $\approx 1/3$ are the "out-of-bag" observations). (iii) For $n = 5$: $\Pr(\text{included}) = 1 - (1 - \frac{1}{5})^5 = 1 - 0.8^5 = 1 - 0.328 = \mathbf{0.672}$.

(b) Bias–variance decomposition.

$$\mathrm{MSE}_1 = \mathrm{Var} + \mathrm{Bias}^2 + \mathrm{Var}(\varepsilon) = 3.5 + (-1.0)^2 + 2.0 = 3.5 + 1.0 + 2.0 = \mathbf{6.5},$$

$$\mathrm{MSE}_2 = 0.5 + (-2.5)^2 + 2.0 = 0.5 + 6.25 + 2.0 = \mathbf{8.75}.$$

Deploy the **flexible spline** (method 1): its higher estimation variance (3.5 vs. 0.5) is more than offset by its much smaller squared bias (1.0 vs. 6.25), giving the lower expected test error ($6.5 < 8.75$). The lowest test MSE any method can achieve at x_0 is the *irreducible error* $\mathrm{Var}(\varepsilon) = \mathbf{2.0}$; no estimator can go below it.

(c) Elastic net. The elastic net minimises, for $0 \le \alpha \le 1$,

$$\sum_{i=1}^{n}\Big(y_i - \beta_0 - \textstyle\sum_j \beta_j x_{ij}\Big)^2 + \lambda\Big[\tfrac{1-\alpha}{2}\textstyle\sum_j \beta_j^2 + \alpha \sum_j |\beta_j|\Big],$$

i.e. a weighted mix of the ridge (ℓ_2) and lasso (ℓ_1) penalties. From the *lasso* it inherits the ability to set coefficients *exactly to zero* (sparsity / variable selection); from *ridge* it inherits the graceful handling of correlated predictors and the ability to select more than n variables when $p > n$. It clearly beats the lasso when there are *groups of highly correlated predictors*: the lasso tends to pick one member of a group essentially at random and drop the rest, whereas the elastic net's ridge part shares the weight across the group and keeps them in or out together (the "grouping effect").

(d) Collinearity and VIF. Collinearity does not bias the OLS estimates, but it makes them *imprecise and unstable*: it inflates their variances and standard errors, widens confidence intervals, and can make individual coefficients change sign or lose significance even when the predictors are jointly important (predictions may still be fine). With $R_j^2 = 0.9$,

$$\mathrm{VIF}_j = \frac{1}{1 - 0.9} = \frac{1}{0.1} = \mathbf{10},$$

so the variance of $\hat{\beta}_j$ is 10 times larger than it would be with an uncorrelated X_j, and its standard error is inflated by $\sqrt{10} = \mathbf{3.162}$. As a rule of thumb a VIF above 5–10 signals problematic collinearity.

(e) Matching.

- (i) **Linear regression**: quantitative response, goal is *inference* (effect size with confidence interval), $n \gg p$ — exactly what OLS with standard errors provides.
- (ii) **Lasso**: $p \gg n$ (8,000 vs. 80) makes OLS infeasible; the ℓ_1 penalty regularises and selects a short list of metabolites.
- (iii) **Logistic regression**: binary response with interpretable coefficients ($e^{\hat{\beta}_j}$ = odds ratios) — suitable for reporting how each characteristic changes the odds.
- (iv) **Random forest**: captures non-linearities and interactions automatically from many predictors and is accurate and robust; interpretability is not required, only forecast accuracy.

End of the examination — good luck!